

# Using Sentiment Analysis to Evaluate Hockey Player Performance

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**Abstract**—Individual hockey player performance is difficult to evaluate using only traditional box scores, as on-ice impact such as positioning, effort, and physicality is hard to quantify. This project aims to supplement player statistics through sentiment analysis of Reddit game thread discussions, classifying viewer comments that mention specific players as positive, neutral, or negative. The NLP pipeline scrapes all top-level comments from a Reddit game thread, uses NHL API data to identify active players, matches and filters for comments mentioning active players, performs sentiment analysis classification on the comments, and aggregates sentiment metrics for each player. Rule-based and transformer-based sentiment analysis approaches are evaluated, with a fine-tuned RoBERTa model achieving the best performance at 0.76 accuracy and a 0.74 Macro F1-score. In an end-to-end test run, a comparison of player sentiment score and plus-minus showed a generally positive relationship, suggesting that the sentiment score captures some patterns consistent with traditional player performance indicators.

**Index Terms**—Sentiment analysis, opinion mining, social media analysis, natural language processing

## I. INTRODUCTION

Ice hockey is a sport with continuous gameplay, where players constantly swap in and out of the game with few stoppages in play. Each player typically remains on the ice for short shifts of one to two minutes before being replaced, resulting in limited opportunities to accumulate statistics that are widely tracked, such as goals, assists, or hits. Unlike sports like basketball or baseball, where player performance can be summarized entirely from their box score data, an ice hockey player could play very well in twenty minutes of ice time without registering any meaningful statistics. During a hockey game, standard box scores only include basic data that do not fully capture the impact of a player in the game. Players could be a positive factor in the game through measures that are difficult to quantify such as playing shutdown defense, winning board battles, or generally being hard to play against. On the other hand, bad positioning, lack of effort, and poor passing are all examples of negative play that do not appear in the box score. As such, it can be difficult to judge how well a hockey player is performing without watching the entire 60 minute game.

With the available statistics not fully capturing an individual hockey player's on-ice impact in a given game, a potential solution that is independent of statistics could come from sentiment analysis. Sentiment analysis is the process

of computationally analyzing a piece of text or media to determine whether the author's opinion on a particular subject is positive, negative, or neutral [1], [2]. This technique has been applied across many different web intelligence domains, including sports analysis, social media troll account detection, and political discussion [4]–[7]. Using sentiment analysis models, we can approximate the perceived performance of hockey players in a game, aggregating real-time fan reactions to their play through natural language processing of social media comments.

The goal of this project is to provide another way to quickly assess hockey player performance to supplement the traditional box score. During major hockey matchups, such as NHL or Olympic games, thousands of fans share live reactions and opinions on social media platforms, with Reddit being one of the most popular platforms for live discussion. In this project, we scrape comments that mention specific players from live game threads posted in the team subreddit, exploring both rule-based and transformer-based sentiment analysis approaches to produce an aggregate sentiment analysis score that represents the fanbase perception of a player's performance in a given game. This project is relevant to Web Intelligence because it analyzes large amounts of user-generated social media content and transforms unstructured online discussion into structured, player-level sentiment metrics.

The contributions of this project are as follows:

- A script that scrapes Reddit threads, collecting all top-level comments.
- Player matching for nicknames and common aliases, and redaction of names for unbiased model training.
- A manually labeled dataset of Reddit comments and their sentiment, used for training and evaluation.
- Evaluation and comparison of rule-based and transformer-based sentiment analysis approaches.
- An end-to-end NLP pipeline that scrapes Reddit comments and performs sentiment analysis on them, producing a centralized table with sentiment metrics and traditional statistics, enabled by the public NHL API.

This project builds on an earlier project proposal written for CS 200W (Graduate Technical Writing) [3]. The actual implementation, experiments, and results were all completed for CS 274.

The remainder of the report is structured as follows: Section II surveys previous work using social media for sentiment analysis applications. Section III details the proposed NLP pipeline and sentiment analysis models evaluated. Section IV presents model evaluation results and analysis of a full pipeline test run. Section V concludes with future extensions.

## II. RELATED WORK

Previous research has demonstrated that sentiment analysis can be effectively applied to social media discourse to evaluate sports performance. Li et al. scraped Twitter for posts that mentioned specific basketball players by name during the NBA playoffs, then performed sentiment analysis to assess public opinion of individual player performance [4]. For their experiments, they used the TextBlob rule-based Python library to generate polarity and subjectivity scores, using these scores to conduct a correlation analysis with performance metrics derived from the actual box score statistics of the players. In this project, sentiment analysis is applied to Reddit comments instead of Twitter posts, as the rise in popularity of sports betting in recent years has led to additional bias on Twitter, where many gamblers complain about players who perform well overall but fail to meet specific betting thresholds. Live discussion in the game threads posted in team-specific subreddits consists primarily of dedicated fans and frequent viewers, leading to a more representative assessment of player performance.

Beyond sports, researchers have used sentiment analysis to study user behavior in niche online communities. Ni et al. explored the rise of rude and aggressive language on online social platforms, specifically focusing on "trolling" behavior, which involves intentionally posting provoking responses to elicit an angry response [5]. Data from the Chinese mobile game forum NGABBS was analyzed, using NLP models such as Naive Bayes, SimpleRNN, and LSTM to perform sentiment analysis. Their results indicated that deep learning architectures outperform traditional models, with their LSTM model achieving the highest performance. Similarly, Miao et al. built a meta-learning model for the early detection of multilingual troll accounts, aiming to combat troll account farms used to spread misinformation or manipulate public opinion [6]. In their evaluations, a multilingual BERT-based model outperformed other models on evaluation datasets, demonstrating the effectiveness of transformer models in identifying social media behaviors and motivating the use of BERT-based models in our experiments.

De la Peña et al. applied sentiment analysis and topic modeling methods to Reddit and Instagram discussions regarding the U.S. Department of Education [7]. They used both VADER (Valence Aware Dictionary and sEntiment Reasoner) and BERT for sentiment analysis, discovering that Reddit was more likely to contain negative sentiment compared to more balanced discourse on Instagram. To model the relationship between sentiment and engagement, the researchers used correlation and regression techniques, but did not find any meaningful relationship. Their work supports the use of both

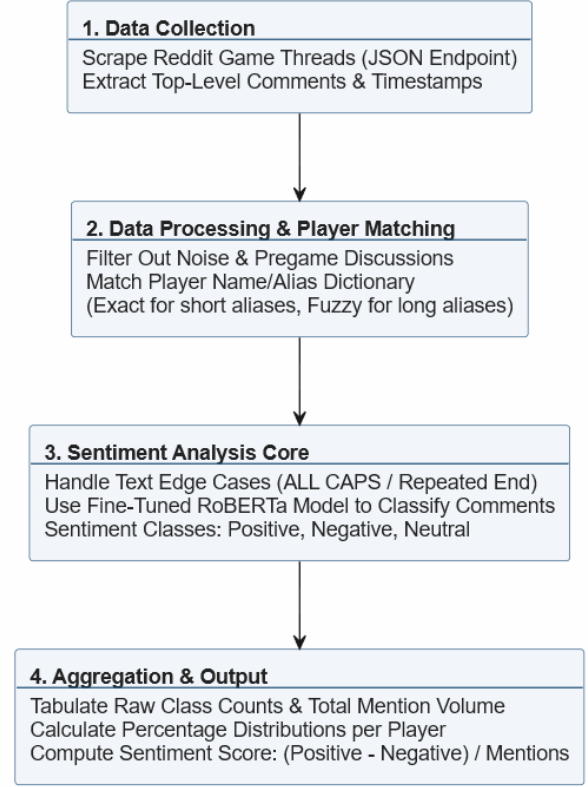


Fig. 1. System architecture for the Reddit sentiment analysis pipeline.

rule-based and transformer-based approaches for analyzing online social media posts.

Building on these previous works, this project focuses on Reddit hockey game threads to reduce the noise observed in Twitter-based studies, and both rule-based and transformer-based approaches are explored for sentiment analysis of player-related comments.

## III. METHODOLOGY

The overall pipeline of this project consists of data collection, data processing, sentiment classification, and aggregation. First, comment data are scraped from Reddit game threads after a game concludes, and only comments that mention active players in the game are retained. The processed text is passed into a RoBERTa sentiment analysis model that has been fine-tuned for this specific domain, which classifies each comment as positive, negative, or neutral. Finally, the sentiment predictions are aggregated to produce a sentiment summary for each player that reflects viewer perception of performance in a given game. This system architecture is summarized in Figure 1.

### A. Data Collection

To collect the Reddit data, we wrote a Python script that scrapes comments from a specified Reddit game thread using Reddit's JSON endpoint, with only top-level comments

retained to focus on primary user reactions and reduce noise from nested discussions.

For each comment, the script extracts the raw text content and timestamp, and stores the data in a CSV file for further processing. Ideally, this script is run after the completion of a game to capture all fan reactions over the course of the entire game.

### B. Data Processing and Player Matching

After collecting the comments from the Reddit game thread, we filter out pregame discussion such as predictions or hype that happens before the game starts, as these comments do not reflect in-game player performance.

For this project, we focused on comments related to the San Jose Sharks to simplify player identification. We constructed a lookup dictionary for the Sharks roster consisting of common nicknames and variations of player names used in Reddit comments.

Each comment is split into individual words and compared against the alias dictionary to identify player mentions. Shorter aliases of four or less characters require exact matches to reduce false positives, while fuzzy string matching is applied for longer aliases to handle misspellings and informal variations. Comments containing at least one matched player are retained, and these comments along with the detected player names and matched aliases are stored in a dataframe for later sentiment analysis. A single comment might mention multiple players, in which case each mentioned player receives the sentiment value of that comment.

### C. Sentiment Analysis

1) *Final Sentiment Classification Approach:* The dataframe of comments that mention players is passed into a fine-tuned RoBERTa model, which classifies each comment as positive, neutral, or negative sentiment, along with a confidence score [8]. Before applying the model, we handle edge cases where the comment only consists of a player alias in all caps (ex. ASKAROV) or with the last letter repeated (ex. eklunddddddd), labeling them as positive as these comments indicate excitement.

2) *Sentiment Analysis Model Evaluation:* The sentiment analysis model is the core component of this project, and we evaluated several models before selecting the best-performing model for the final implementation.

We implemented the base VADER sentiment analysis model without additional preprocessing of comments as the sentiment analysis baseline. VADER is a rule-based model designed for social media text, making it well-suited for Reddit comments [9]. This model returns a compound sentiment strength score ranging from -1 to 1. In our evaluations, we used the standard thresholds of compound scores greater than or equal to 0.05 for positive sentiment, compound scores less than or equal to -0.05 for negative, and the remaining scores as neutral. To improve performance on hockey-related content, we also tested extending the VADER lexicon with domain-specific terminology (for example, “clutch”, “liability”, “sniped”).

TABLE I  
EVALUATION DATASET SENTIMENT CLASSES

| Class Label  | Count      | Percentage    |
|--------------|------------|---------------|
| Positive     | 171        | 42.0%         |
| Negative     | 128        | 31.5%         |
| Neutral      | 108        | 26.5%         |
| <b>Total</b> | <b>407</b> | <b>100.0%</b> |

For comparison, we also evaluated TextBlob, which was used in a similar study by Li et al. on NBA-related Twitter posts [4], [10]. TextBlob is a general-purpose text processing library that provides rule-based sentiment analysis designed for general English. We used the PatternAnalyzer implementation, setting the polarity thresholds as greater than 0.1 for positive sentiment, less than -0.1 for negative sentiment, and neutral sentiment in between.

RoBERTa is a BERT-based model that improves on BERT through increased training and new pretraining objectives [8]. We chose to use the Twitter-RoBERTa-base for Sentiment Analysis model for a transformer-based approach, as this best matches the Reddit comment input [11].

As a follow-up, we fine-tuned the RoBERTa model on hockey Reddit comments to improve its handling of hockey-specific terminology and slang. Before training, player names were redacted to prevent the model from overfitting to specific names.

For training and evaluation purposes, we constructed a dataset of 407 comments mentioning specific players and manually labeled each comment as positive, neutral, or negative sentiment. The class distribution of this dataset is shown in Table I. The dataset is somewhat unbalanced, with positive examples making up about 42% of comments. VADER, TextBlob, and base RoBERTa were evaluated on the full constructed labeled dataset to compare their performance in classifying hockey-related sentiment. The RoBERTa model was fine-tuned and evaluated using a stratified 85/15 train-validation split of the labeled dataset.

Sentiment analysis models were quantitatively evaluated using accuracy to measure overall classification performance and precision, recall, and F1-score for performance across the three sentiment classes. Models were qualitatively evaluated by examining misclassified comments.

### D. Aggregation and Output

The final stage of the pipeline involves aggregating sentiment classifications at the player level to produce a sentiment summary. The public NHL API provides access to box scores for every NHL game, containing information such as game start time, team rosters, and player statistics [12]. The active roster for a specific game is pulled using the NHL API, and for each player on the roster, the number of positive, neutral, and negative labeled comments is tabulated and converted into percentages. The total number of player mentions is also included to provide context for the reliability of the scores.

TABLE II  
CLASSIFICATION REPORT: VADER BASELINE

| Class        | Precision | Recall | F1-Score | Support |
|--------------|-----------|--------|----------|---------|
| Negative     | 0.55      | 0.49   | 0.52     | 128     |
| Neutral      | 0.28      | 0.41   | 0.33     | 108     |
| Positive     | 0.52      | 0.42   | 0.46     | 171     |
| Accuracy     |           |        | 0.44     | 407     |
| Macro Avg    | 0.45      | 0.44   | 0.44     | 407     |
| Weighted Avg | 0.47      | 0.44   | 0.45     | 407     |

To summarize the overall sentiment for a player, we defined a sentiment score ranging from -1 to 1:

$$\text{sentiment\_score} = \frac{N_{\text{positive}} - N_{\text{negative}}}{N_{\text{mentions}}} \quad (1)$$

A score closer to 1 indicates mostly positive comments, while a score closer to -1 indicates mostly negative comments. Scores near 0 indicate either mixed sentiment or a large proportion of neutral comments.

These sentiment metrics are joined with the box score as the final output, resulting in a centralized table that contextualizes traditional statistics with viewer perception.

## IV. RESULTS AND DISCUSSION

### A. Sentiment Analysis Model Evaluation

This section evaluates the performance of various sentiment analysis models, which is the foundation for computing sentiment scores for players. In total, we evaluated five configurations: zero-shot VADER, TextBlob, VADER extended with hockey-specific lexicon, zero-shot RoBERTa, and RoBERTa with fine-tuning.

1) *VADER Baseline*: For the baseline experiment, we evaluated the VADER sentiment analysis model on the manually labeled dataset of 407 player-related comments without additional preprocessing or comment cleaning. The full classification report is shown in Table II.

The results indicate that the baseline performs poorly overall, with an accuracy of 0.44 and a macro F1-score of 0.44. The neutral class proves to be the most challenging, producing an F1-score of 0.33, which suggests that VADER struggles to distinguish neutral player evaluations from clearly positive or negative reactions. The negative and positive classes perform somewhat better, with F1-scores of 0.52 and 0.46 respectively, but these results are still not satisfactory.

Qualitatively inspecting the misclassified comments, these errors are often caused by sarcasm, domain-specific hockey terminology, and context-dependent profanity, all of which are difficult for rule-based models to interpret accurately.

2) *TextBlob*: We evaluated the TextBlob sentiment analysis model for comparison, used in the reference paper by Li et al. [4].

As shown in Table III, the TextBlob model achieves an overall accuracy of 0.43 and a macro F1-score of 0.43, which is slightly lower than the VADER baseline. Similarly to VADER, TextBlob struggles across all classes, with the neutral

TABLE III  
CLASSIFICATION REPORT: TEXTBLOB MODEL

| Class        | Precision | Recall | F1-Score | Support |
|--------------|-----------|--------|----------|---------|
| Negative     | 0.53      | 0.42   | 0.47     | 128     |
| Neutral      | 0.30      | 0.51   | 0.38     | 108     |
| Positive     | 0.53      | 0.37   | 0.44     | 171     |
| Accuracy     |           |        | 0.43     | 407     |
| Macro Avg    | 0.45      | 0.44   | 0.43     | 407     |
| Weighted Avg | 0.47      | 0.43   | 0.43     | 407     |

TABLE IV  
CLASSIFICATION REPORT: VADER WITH AUGMENTATION

| Class        | Precision | Recall | F1-Score | Support |
|--------------|-----------|--------|----------|---------|
| Negative     | 0.56      | 0.52   | 0.54     | 128     |
| Neutral      | 0.30      | 0.41   | 0.35     | 108     |
| Positive     | 0.53      | 0.45   | 0.49     | 171     |
| Accuracy     |           |        | 0.46     | 407     |
| Macro Avg    | 0.47      | 0.46   | 0.46     | 407     |
| Weighted Avg | 0.48      | 0.46   | 0.47     | 407     |

class again producing the lowest F1-score at 0.38. The positive and negative classes achieve F1-scores of 0.44 and 0.47, respectively, both of which are lower than the corresponding VADER scores. This is not a surprising result, as VADER is optimized for social media text, while TextBlob is designed for general English.

3) *VADER with Augmentation*: To improve the VADER model, we updated the lexicon weights to incorporate hockey-specific language. The classification report is shown in Table IV.

The augmented VADER model achieves an overall accuracy of 0.46 and a macro F1-score of 0.46, which represents an improvement of 0.02 in both accuracy and macro F1-score over the baseline VADER model. Performance improves marginally across all classes, with the F1-score for the neutral class increasing from 0.33 to 0.35, the negative class from 0.52 to 0.54, and the positive class from 0.46 to 0.49.

Incorporating hockey-specific language into the VADER lexicon does improve sentiment classification performance, but the overall performance remains relatively low. The results indicate that challenges such as sarcasm, context-dependent profanity, and multi-player references are not fully addressed by lexicon-based approaches alone.

4) *RoBERTa Baseline*: The zero-shot RoBERTa model shows a significant improvement compared to the augmented VADER model, with a 0.17 increase in both macro F1-score and accuracy, as seen in Table V. The neutral class remains the most challenging class, with RoBERTa producing an F1-score of 0.47, which is still an improvement over the 0.34 F1-score achieved by the augmented VADER model.

Even without optimization for hockey-related language, the RoBERTa model outperformed the rule-based sentiment analysis models, showing the effectiveness of transformer models for incorporating additional context for sentiment analysis. The model's pretraining on Twitter data allows it to better deal

TABLE V  
CLASSIFICATION REPORT: ROBERTA BASELINE

| Class        | Precision | Recall | F1-Score | Support |
|--------------|-----------|--------|----------|---------|
| Negative     | 0.67      | 0.77   | 0.72     | 128     |
| Neutral      | 0.43      | 0.53   | 0.47     | 108     |
| Positive     | 0.82      | 0.60   | 0.69     | 171     |
| Accuracy     |           |        | 0.63     | 407     |
| Macro Avg    | 0.64      | 0.63   | 0.63     | 407     |
| Weighted Avg | 0.67      | 0.63   | 0.64     | 407     |

TABLE VI  
CLASSIFICATION REPORT: FINE-TUNED ROBERTA MODEL

| Class        | Precision | Recall | F1-Score | Support |
|--------------|-----------|--------|----------|---------|
| Negative     | 0.83      | 0.75   | 0.79     | 20      |
| Neutral      | 0.64      | 0.56   | 0.60     | 16      |
| Positive     | 0.77      | 0.88   | 0.82     | 26      |
| Accuracy     |           |        | 0.76     | 62      |
| Macro Avg    | 0.75      | 0.73   | 0.74     | 62      |
| Weighted Avg | 0.76      | 0.76   | 0.77     | 62      |

with internet lingo, sarcasm, and short comments, but it still struggles with hockey-specific language and with profanity, often labeling comments as negative when commenters use profanity out of excitement.

5) *RoBERTa with Fine-Tuning*: Compared to the zero-shot model, the fine-tuned RoBERTa model showed a 0.14 improvement in accuracy to 0.76 as shown in Table VI, indicating that fine-tuning on Reddit comments helped the model better understand the hockey-related language and expressions observed in game thread discussions. The fine-tuned model also greatly outperforms the rule-based sentiment analysis models TextBlob and VADER (augmented), which showed accuracies of 0.43 and 0.46, respectively. With a Macro F1 of 0.74, it shows an improved, more balanced performance across the three classes.

The confusion matrix in Figure 2 shows that the model is able to classify most positive and negative comments correctly, but continues to struggle with the neutral class (recall 0.56), mislabeling 5 of 16 neutral comments as positive and 2 of 16 as negative.

As the fine-tuned RoBERTa model was the best-performing model across all metrics, we chose to use this model in the final sentiment analysis pipeline.

A current limitation of this fine-tuning experiment is that the training dataset is still fairly small at 345 samples and is somewhat imbalanced in class distribution. The validation dataset, which is a 15% split of the labeled dataset, only contains 62 samples. To address this, we would need to manually label more Reddit comments to create a larger and more balanced dataset.

### B. End-to-End Pipeline Output

A test run of the full pipeline was performed using comments from a Reddit game thread in r/SanJoseSharks for a game between the San Jose Sharks and Nashville Predators on

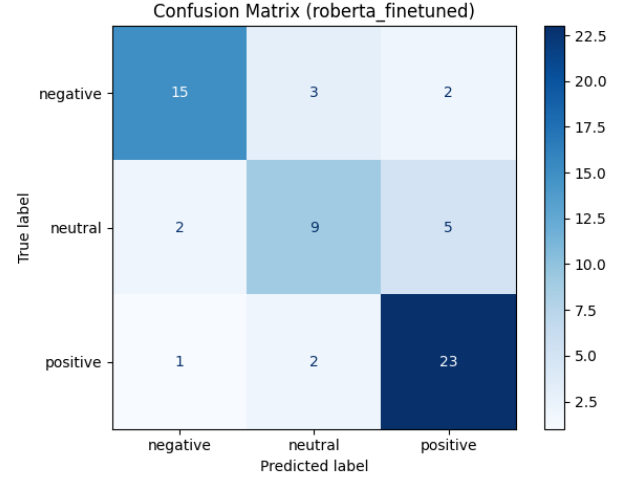


Fig. 2. Confusion matrix for the fine-tuned RoBERTa model.

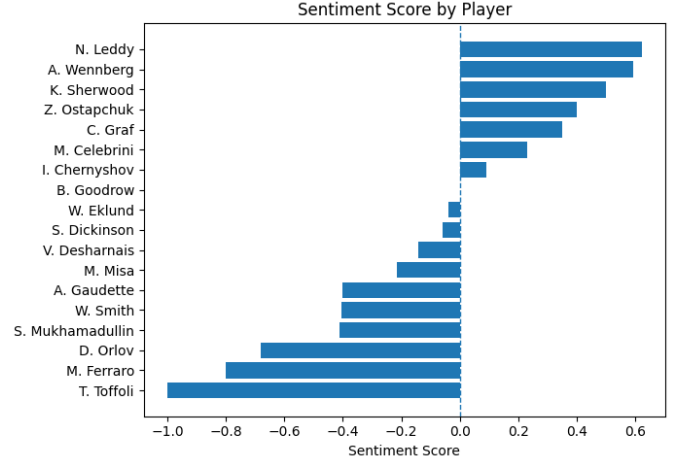


Fig. 3. Player-level sentiment scores calculated for the San Jose Sharks skaters.

April 4, 2026 [13]. The centralized table containing both sentiment metrics and traditional box score statistics is included in Appendix A (Figure 6).

Figure 3 shows the sentiment scores that were calculated from the Reddit comments for the Sharks' skaters in this game. Leddy, Wennberg, and Sherwood received the most positive sentiment, and Toffoli, Ferraro, and Orlov received the most negative. However, looking at Figure 4, which shows the sentiment distribution and number of mentions for each player, we see that Toffoli and Ferraro were among the least mentioned players, making their scores less reliable. In contrast, Will Smith had by far the highest number of negatively labeled comments, which suggests that the negative sentiment towards him is strongly supported with a large number of viewers reacting negatively in comments mentioning him.

To compare how the sentiment score aligns with a traditional statistic, we plot the sentiment score against skater plus-minus, displayed in Figure 5. The plus-minus statistic

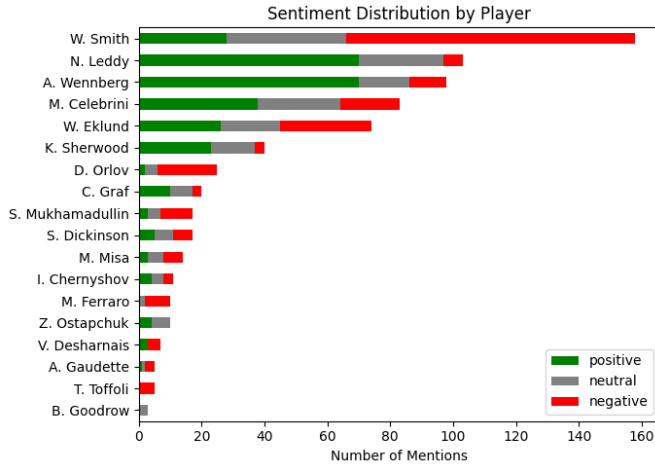


Fig. 4. Distribution of positive, neutral, and negative labeled comments for each San Jose Sharks skater mentioned in the Reddit game thread.

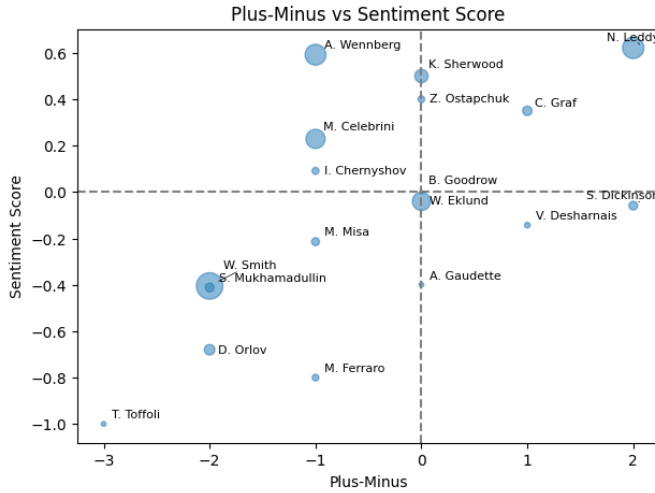


Fig. 5. Comparison of plus-minus and Reddit sentiment score for San Jose Sharks skaters. Point size represents the number of Reddit comments mentioning each player.

captures the even-strength and shorthanded goal differential while a player is on the ice, serving as a rough measure of individual performance. The scatter plot indicates that there generally appears to be a positive relationship between plus-minus and sentiment score, which suggests that the sentiment score is able to capture some patterns that are consistent with traditional player performance indicators. This result should be interpreted with caution, as it is based on a single game.

### C. Discussion

The main strength of this approach is that it provides a supplement to the traditional box score by incorporating fan reactions and contextual information that may not appear in standard statistics. The pipeline approximates how dedicated fans perceived each player's performance during a specific game, producing a player-level sentiment score. The system

has a relatively low runtime, making it a possible candidate for real-time sentiment analysis.

However, there are several limitations in the current implementation. Reddit users are biased toward the most popular and disliked players, with those players receiving a disproportionate number of comments. Sentiment analysis reports for players with few mentions are likely unreliable.

The current best-performing sentiment analysis model, fine-tuned RoBERTa, achieved an accuracy of 0.76 and a Macro F1-score of 0.74. This model was able to mostly overcome the issues with profanity and sarcasm that other models faced, but it still struggles with short ambiguous comments and hockey-specific context. The misclassified comments are included in Appendix B (Table VII). For more reliable results, we would need to expand the manually labeled dataset with more examples.

Additionally, further preprocessing of comments may also improve player sentiment scores. Common issues are players sharing the same name, comments that are not game-specific, and comments that mention multiple players. A limitation of the player recognition system is that it incorrectly flags comments for players with names or nicknames that appear in the English language, such as "Will". The system also has no way to separate groups mentioned in the comments, such as "the Celebrini line". In this example, the sentiment label would only be applied to Celebrini, with his linemates going undetected.

## V. CONCLUSION

This project presented a new approach for assessing hockey player performance by applying sentiment analysis to Reddit game thread comments as a complement to traditional box score statistics. The system scrapes Reddit comments, identifies comments that mention players, uses a fine-tuned RoBERTa model for sentiment analysis to label comment sentiment, and aggregates the results into player-level sentiment summaries. The generated sentiment scores provide additional context to the statistics, showing a generally positive relationship with skater plus-minus for the end-to-end test run.

Evaluation of sentiment analysis models resulted in a RoBERTa model fine-tuned on Reddit comment data achieving the best performance across all metrics, with an accuracy of 0.76 and Macro F1-score of 0.74. The neutral class was the most challenging class to identify, with a low recall of 0.56. Short ambiguous comments and hockey terminology still prove to be difficult for the model.

Future work could improve the system by expanding the manually labeled dataset for model training, testing the pipeline across more teams and games, and improving preprocessing for comments that mention multiple players or contain ambiguous player references. The system could also explore incorporating temporal context from game events to help classify comments more accurately. Reddit comment upvotes could be used as weights when aggregating player sentiment to give more influence to the opinions that other people agreed with. In a real-time application, timestamps

from Reddit comments could be synced with game events to identify when fans reacted to specific plays, allowing users to pinpoint the moments that contributed to a player's sentiment score.

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# APPENDIX A PIPELINE CENTRALIZED BOX SCORE OUTPUT

|    | player           | positive | neutral | negative | mentions | positive_pct | neutral_pct | negative_pct | sentiment_score | plus_minus | goals | assists | points | shots | toi   | toi_minutes |
|----|------------------|----------|---------|----------|----------|--------------|-------------|--------------|-----------------|------------|-------|---------|--------|-------|-------|-------------|
| 0  | W. Smith         | 28       | 38      | 92       | 158      | 0.177215     | 0.240506    | 0.582278     | -0.405063       | -2         | 0     | 0       | 0      | 3     | 18:52 | 18.866667   |
| 1  | N. Leddy         | 70       | 27      | 6        | 103      | 0.679612     | 0.262136    | 0.058252     | 0.621359        | 2          | 1     | 1       | 2      | 2     | 20:42 | 20.700000   |
| 2  | A. Wennberg      | 70       | 16      | 12       | 98       | 0.714286     | 0.163265    | 0.122449     | 0.591837        | -1         | 1     | 0       | 1      | 2     | 20:31 | 20.516667   |
| 3  | M. Celebrini     | 38       | 26      | 19       | 83       | 0.457831     | 0.313253    | 0.228916     | 0.228916        | -1         | 1     | 0       | 1      | 5     | 23:40 | 23.666667   |
| 4  | W. Eklund        | 26       | 19      | 29       | 74       | 0.351351     | 0.256757    | 0.391892     | -0.040541       | 0          | 0     | 1       | 1      | 1     | 14:49 | 14.816667   |
| 5  | K. Sherwood      | 23       | 14      | 3        | 40       | 0.575000     | 0.350000    | 0.075000     | 0.500000        | 0          | 0     | 0       | 0      | 1     | 13:00 | 13.000000   |
| 6  | D. Orlov         | 2        | 4       | 19       | 25       | 0.080000     | 0.160000    | 0.760000     | -0.680000       | -2         | 0     | 1       | 1      | 3     | 22:41 | 22.683333   |
| 7  | C. Graf          | 10       | 7       | 3        | 20       | 0.500000     | 0.350000    | 0.150000     | 0.350000        | 1          | 0     | 1       | 1      | 3     | 17:03 | 17.050000   |
| 8  | S. Mukhamadullin | 3        | 4       | 10       | 17       | 0.176471     | 0.235294    | 0.588235     | -0.411765       | -2         | 0     | 0       | 0      | 0     | 13:02 | 13.033333   |
| 9  | S. Dickinson     | 5        | 6       | 6        | 17       | 0.294118     | 0.352941    | 0.352941     | -0.058824       | 2          | 0     | 0       | 0      | 2     | 20:58 | 20.966667   |
| 10 | M. Misa          | 3        | 5       | 6        | 14       | 0.214286     | 0.357143    | 0.428571     | -0.214286       | -1         | 0     | 0       | 0      | 0     | 11:10 | 11.166667   |
| 11 | I. Chernyshov    | 4        | 4       | 3        | 11       | 0.363636     | 0.363636    | 0.272727     | 0.090909        | -1         | 0     | 0       | 0      | 2     | 14:15 | 14.250000   |
| 12 | M. Ferraro       | 0        | 2       | 8        | 10       | 0.000000     | 0.200000    | 0.800000     | -0.800000       | -1         | 0     | 0       | 0      | 0     | 16:34 | 16.566667   |
| 13 | Z. Ostapchuk     | 4        | 6       | 0        | 10       | 0.400000     | 0.600000    | 0.000000     | 0.400000        | 0          | 0     | 0       | 0      | 0     | 10:09 | 10.150000   |
| 14 | V. Desharnais    | 3        | 0       | 4        | 7        | 0.428571     | 0.000000    | 0.571429     | -0.142857       | 1          | 0     | 0       | 0      | 1     | 20:44 | 20.733333   |
| 15 | T. Toffoli       | 0        | 0       | 5        | 5        | 0.000000     | 0.000000    | 1.000000     | -1.000000       | -3         | 0     | 0       | 0      | 2     | 14:52 | 14.866667   |
| 16 | A. Gaudette      | 1        | 1       | 3        | 5        | 0.200000     | 0.200000    | 0.600000     | -0.400000       | 0          | 0     | 0       | 0      | 0     | 09:39 | 9.650000    |
| 17 | B. Goodrow       | 0        | 3       | 0        | 3        | 0.000000     | 1.000000    | 0.000000     | 0.000000        | 0          | 0     | 0       | 0      | 0     | 10:49 | 10.816667   |

Fig. 6. Box score statistics and Reddit sentiment results for San Jose Sharks skaters in the April 4, 2026 game against the Nashville Predators.

## APPENDIX B ROBERTA (FINE-TUNED) MISCLASSIFIED COMMENTS

TABLE VII  
MISCLASSIFIED COMMENTS FROM THE FINE-TUNED ROBERTA MODEL

| Comment                                                                                                                           | True Sentiment | Predicted |
|-----------------------------------------------------------------------------------------------------------------------------------|----------------|-----------|
| [PLAYER] going in slow motion there                                                                                               | Negative       | Neutral   |
| [PLAYER] back on the first. A thing of beauty.                                                                                    | Neutral        | Positive  |
| RIP. It's crazy how high my expectations are from [PLAYER]. I expect perfection from a 19 year old because of how good he's been. | Neutral        | Positive  |
| GET RID OF [PLAYER] [PLAYER]                                                                                                      | Negative       | Neutral   |
| [PLAYER] [PLAYER] is now in 7th for single season points                                                                          | Positive       | Neutral   |
| Paws off [PLAYER]!                                                                                                                | Neutral        | Positive  |
| [PLAYER] gets a shoulder to the head and somehow that that's his fault? what's mcavoy even mad about???                           | Neutral        | Negative  |
| [PLAYER] wants it so badly and so do I need this goal drought to end                                                              | Neutral        | Positive  |
| [PLAYER] heard y'all talkin shit                                                                                                  | Positive       | Negative  |
| Gotta take that shot [PLAYER].                                                                                                    | Negative       | Neutral   |
| Lord blew right by [PLAYER] like he was standing still.                                                                           | Negative       | Positive  |
| If [PLAYER] gained like 10 more lbs of muscle... he'd be unstoppable                                                              | Neutral        | Positive  |
| [PLAYER] ily but u gotta be physical                                                                                              | Negative       | Positive  |
| [PLAYER] needs to be on the pp1                                                                                                   | Positive       | Neutral   |
| the [PLAYER] scoring stat does not apply anymore                                                                                  | Neutral        | Negative  |